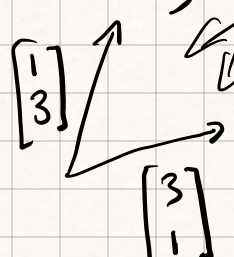


back to ex in $\mathbb{R}^2$ w/ dot product

 $\tilde{b}_1 := \frac{b_1}{\|b_1\|}$ need the part of b_2 that is ⊥ to b_1
which we know is $(b_2 - \text{proj}_{b_1} b_2)$

$$\tilde{b}_2 := \frac{b_2 - \text{proj}_{b_1} b_2}{\|b_2 - \text{proj}_{b_1} b_2\|}$$

Ex: $\mathbb{R}^2$ dot product

 $\tilde{b}_1 = \frac{\begin{bmatrix} 1 \\ 3 \end{bmatrix}}{\sqrt{\begin{bmatrix} 1 \\ 3 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 3 \end{bmatrix}}} = \begin{bmatrix} 1/\sqrt{10} \\ 3/\sqrt{10} \end{bmatrix}$

$$\text{proj}_{b_1} b_2 = \frac{\langle \begin{bmatrix} 1 \\ 3 \end{bmatrix}, \begin{bmatrix} 3 \\ 1 \end{bmatrix} \rangle}{\langle \begin{bmatrix} 1 \\ 3 \end{bmatrix}, \begin{bmatrix} 1 \\ 3 \end{bmatrix} \rangle} \begin{bmatrix} 1 \\ 3 \end{bmatrix} = \frac{6}{10} \begin{bmatrix} 1 \\ 3 \end{bmatrix} = \begin{bmatrix} 6/10 \\ 18/10 \end{bmatrix}$$

$$\begin{aligned} \tilde{b}_2 &= \frac{\begin{bmatrix} 3 \\ 1 \end{bmatrix} - \begin{bmatrix} 6/10 \\ 18/10 \end{bmatrix}}{\|b_2 - \text{proj}_{b_1} b_2\|} = \frac{\begin{bmatrix} 24/10 \\ -8/10 \end{bmatrix}}{\sqrt{\begin{bmatrix} 24/10 \\ -8/10 \end{bmatrix} \cdot \begin{bmatrix} 24/10 \\ -8/10 \end{bmatrix}}} = \frac{\begin{bmatrix} 24/10 \\ -8/10 \end{bmatrix}}{\sqrt{\frac{640}{100}}} = \frac{\begin{bmatrix} 24/10 \\ -8/10 \end{bmatrix}}{8\sqrt{1/10}} \\ &= \begin{bmatrix} 3\sqrt{10}/10 \\ -\sqrt{10}/10 \end{bmatrix} \quad \langle \tilde{b}_1, \tilde{b}_2 \rangle = 3/10 - 3/10 = 0 \quad \checkmark \end{aligned}$$

Gram Schmidt process

1) works for any inner product

2) works in any finite # of dims

$$\tilde{b}_1 = \frac{b_1}{\|b_1\|} \quad \tilde{b}_2 = \frac{b_2 - \text{proj}_{b_1} b_2}{\|b_2 - \text{proj}_{b_1} b_2\|} \quad \tilde{b}_3 = \frac{b_3 - \text{proj}_{b_1} b_3 - \text{proj}_{b_2} b_3}{\| \underbrace{b_3 - \text{proj}_{b_1} b_3 - \text{proj}_{b_2} b_3} \|}$$

Thm: Given any inner product for a finite dim vector space there exists a basis where the formula for the inner product is the dot product

Lem: in any orthonormal basis the inner product is the dot product

Lem: any basis can be turned into an orthonormal basis by the gram schmidt process

note: there is tension abt how to choose a basis

problem $\longleftrightarrow$ inner product-apt
- apt

EX: commodities

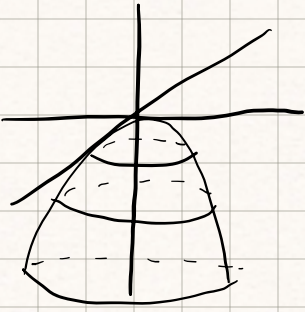
$\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ each coordinate corresponds exactly to a quantity of one commodity

An inner product on this space is a way of assigning "size" the problem-apt coordinates may not be orthonormal knowing how/when to switch coordinates gives the advantages of both

Optimization

given $f: \mathbb{R}^m \rightarrow \mathbb{R}$ we might ask does this func have a global maximum?

ex: $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ $f(x,y) = -(x^2 + y^2)$



visually we see that $(0,0)$ is the only local max and in fact is the global max